

Danison N

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EDUCATION

University of Sydney

Bachelor of Computer Science and Data Science (Physics)

Jan 2023 – Jan 2026

Sydney, NSW

EXPERIENCE

Software Engineer Intern

AiWorkFlo

Sep 2025 – Nov 2025

Sydney, NSW (Hybrid)

- Implemented a custom **interval scheduling algorithm** to reduce booking conflicts by 90% and free **8–15 extra slots/week** per clinic, boosting weekly revenue conversion by **20%**, built on Meta.
- Developed a scalable **Flask-based REST API** integrating Plato (EMR) with third-party CRMs, enabling real-time lead ingestion, webhook handling, and asynchronous marketing workflows with **token-based auth**.
- Refactored backend services using **Docker** for containerization and integrated **Amazon SNS** for event-driven notifications. Improved API responsiveness by 40% through async, and enhanced reliability with **unit tests**.

Software Engineer Intern

Zerimar

Mar 2025 – Jun 2025

Sydney, NSW

- Led development of a real-time subcontractor tracking platform using **Google Maps API**, implementing internal job query endpoints and **Dijkstra's algorithm** for optimal route generation.
- Built a geospatial classification system using **bounding-box filtering**, **Manhattan distance**, and coordinate parsing to auto-assign jobs to subregions (NSW, VIC, SA, WA, NZ), cutting manual triage by **30%**.
- Collaborated with consultants and the CEO to automated tests and CI/CD pipelines using **GitHub Actions** to ensure deployment quality and development efficiency.

Student Machine Learning Researcher

National Taiwan University

Nov 2024 – Feb 2025

Taipei, Taiwan

- Improved leopard cat detection by 20% with **YOLO** and **Label Studio**, enhancing classification accuracy. Distributed with Docker and Github Actions. (NVIDIA Blog)
- Benchmarked microfluidic experiments and collaborated with PhD researchers, then independently built an optimized leopard-cat detection system adopted by local council for wildlife-safe road planning.

PROJECTS

Invoice Overseer | FastAPI, AWS, Tesseract OCR, Redis

Aug 2025 – Ongoing

- Engineered a backend system using **FastAPI**, **AWS Lambda**, and **S3** to ingest and parse vendor invoices (PDF, email) using **Tesseract OCR** and regex-based field extraction.
- Built an anomaly detection engine using **Redis** queues and vendor baselines to flag overcharges, duplicate invoices, and altered bank details, reducing manual invoice review time by over 60%.

Qup App | React (TS), Django, PostgreSQL, Railway (Live Demo)

Oct 2024 – Dec 2024

- Built a minimalist restaurant wait-list web app using **React (TypeScript)** and a **Django REST API**, enabling real-time queue management for walk-in customers.
- Deployed the full-stack app on **Railway**, enabling sub-5s cold starts and CI/CD with GitHub Actions.
- Reduced average customer wait time by **35%** during pilot rollout; validated with **2–3 live restaurant partners** in Sydney.

Construction Project Index | React (Live Demo)

Jul 2024 – Oct 2024

- Built a map-based web app to visualise construction projects across Australia, with project selection and natural language queries via **Gemini API**.

Markdown Collaboration Server | C, POSIX Signals, Epoll (GitHub)

Mar 2024 – Jun 2024

- Built a multi-client Markdown collaboration server in **C** using **POSIX** signals and event-driven I/O, supporting real-time edits (INSERT, DEL, BOLD, etc.), role-based permissions, and autosaving to **doc.md**.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, R, TypeScript

Frameworks/Tools: Flask, Django, Docker, React

Cloud/Infra: AWS (EC2, SNS), PostgreSQL, Firebase, Vercel